Financial Decision Support System (FDSS)

Numerical Methods and Monte Carlo Simulation

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Applied Mathematics, Numerical Analysis & Stochastic Asset Modeling

The Core Problem: Why Retail Calculators Lie

The dangerous mathematical gap between static expectations and volatile capital markets

Conventional Retail Projection

- Assumes invariant annual return: $A(t) = P(1 + r)^t$.
- Projects lifetime wealth as an unrealistic straight line.
- Completely ignores volatility ($\sigma = 0$ assumption).
- Violates Jensen's Inequality in multi-period compounding.

Stochastic Market Reality

- Capital markets fluctuate under continuous Brownian motion.
- Volatility exerts a structural mathematical drag on returns.
- Typical median wealth strictly lags the arithmetic average.
- Retail investors experience severe, unexpected shortfalls.

$$\blacksquare[g(X)] \neq g(\blacksquare[X]) \implies$$ **Deterministic formulas hide downside tail risk.**

Jensen's Inequality: Non-linear compounding under variance invalidates simple average calculations.

Research Hypothesis & Applied Mathematics Approach

Grounding retail financial planning in rigorous numerical analysis and probability theory

"Replacing static scalar estimates with numerical root-finding and continuous stochastic processes equips investors with true distribution-aware risk modeling."

Central Research Hypothesis

1. Analytical Theory

- Formulate capital accumulation using classical ODEs.
- Derive continuous compounding via Euler limits.
- Expose consumer loan APR illusions with EAR.

2. Numerical Optimization

- Overcome polynomial insolvability (Abel–Ruffini).
- Implement Newton–Raphson with analytical derivative.
- Guarantee convergence using Bisection fallback.

3. Stochastic Simulation

- Model continuous prices via Geometric Brownian Motion.
- Derive volatility drag proof using Itô's Lemma.
- Generate 10,000-path Monte Carlo risk distributions.

Theoretical Pillar 1: Time Value & Governing ODEs

Linear accumulation versus infinitesimal autonomous growth in continuous time

Simple Accumulation (Linear ODE)

- Governing ODE: $dA/dt = P \cdot r$, with $A(0) = P$.
- Exact solution: $\boldsymbol{A(t) = P(1 + rt)}$.
- Linear growth with constant slope (Arithmetic Progression).
- Violates subadditivity: $A(t_1 + t_2) < A(t_1)A(t_2)/P$.

Continuous Compounding (Autonomous ODE)

- Governing ODE: $dA/dt = r \cdot A(t)$, with $A(0) = P$.
- Exact solution: $\boldsymbol{A(t) = P\,e^{rt}}$.
- Growth rate is directly proportional to accumulated wealth.
- Infinitesimal Euler limit: $\lim_{m\to\infty} (1 + r/m)^{mt} = e^{rt}$.

$$\mathbf{EAR = (1 + r_{nom}/m)^{m} - 1 \quad \bullet \quad r_c = m \cdot \ln(1 + r_d/m)}$$

Key Insight: Continuously compounded logarithmic returns add linearly across time periods.

Theoretical Pillar 2: Consumer Debt Analytics

Finite geometric series summation and the negative amortization boundary theorem

$$P = \sum_{k=1}^{n} E(1+R)^{-k} \quad \Rightarrow \quad E = P \cdot R \cdot (1+R)^{n} \,/\, [\,(1+R)^{n} - 1\,]$$

Equated Monthly Installment (EMI) derived via closed-form inversion of a finite geometric progression ($R = r/12$).

Amortization Dynamics

- Monthly interest: $I_k = B_{k-1} \cdot R$.
- Principal repaid: $P_k = E - I_k$.
- Principal sequence forms an increasing geometric progression.
- Early prepayments collapse total interest exponentially.

Negative Amortization Boundary

- Boundary condition: Loan amortizes $\Leftrightarrow E > P \cdot r / 12$.
- If $E \le P \cdot R$, principal repaid is zero or negative.
- Debt balance diverges to infinity: $\lim_{n \to \infty} B_n = +\infty$.
- Exposes predatory 'minimum payment' credit card traps.

Theoretical Pillar 3: Numerical Yield Optimization

Why Galois theory mandates numerical analysis for multi-period financial yields

$$\text{NPV}(r) = \sum_{t=0}^{T} C_t / (1 + r)^t = 0 \quad \Leftrightarrow \quad \sum_{t=0}^{T} C_t x^t = 0 \quad (x = (1+r)^{-1})$$

The Internal Rate of Return (IRR) polynomial of degree T.

Mathematical Insolvability

- **The Abel–Ruffini Theorem (Galois Theory):**
- No general radical solution exists for degree $T \geq 5$.
- Multi-period financial cash flows cannot be solved algebraically.
- Numerical analysis is fundamentally mathematically mandatory.

Dual-Engine Numerical Solver

- **Newton–Raphson with Analytical Derivative:**
- Iteration: $r_{k+1} = r_k - [\ \text{NPV}(r_k) / f'(r_k)\ ]$.
- Quadratic convergence rate: error squares at each step $|e_{k+1}| \leq M\,|e_k|^2$.
- Bisection fallback guarantees convergence if derivatives oscillate.

Theoretical Pillar 4: Stochastic Asset Dynamics

Continuous-time Brownian motion and the mathematical proof of volatility drag

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t \quad \Rightarrow \quad d(\ln S_t) = (\mu - \tfrac{1}{2}\sigma^2)\,dt + \sigma\,dW_t$$

Itô's Lemma Proof: Non-zero quadratic variation $(dW_t)^2 = dt$ creates structural drag $-\frac{1}{2}\sigma^2$.

Itô's Stochastic Calculus

- In standard calculus, $d(\ln S) = dS/S$.
- Brownian motion has non-zero variation: $(dW)^2 = dt$.
- Itô Taylor expansion includes second derivative: $f''(S) = -1/S^2$.
- Generates exact discrete solution: $S_t = S_0 \exp[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t]$.

The Volatility Drag Divergence

- Arithmetic Mean: $\blacksquare[S_t] = S_0 e^{\mu t}$ (pulled by right skew).
- Typical Median: $S_{med}(t) = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t}$.
- Volatility Drag penalty: $\Delta\mu = -\frac{1}{2}\sigma^2$ per year.
- More than 50% of investors strictly underperform advertised average!

Theoretical Pillar 5: Monte Carlo Simulation & Risk

Vectorized stochastic ensembles, Central Limit Theorem bounds, and Value at Risk

$$\text{Standard Error} = \sigma_{\text{sample}} / \sqrt{M} = \blacksquare(1 / \sqrt{M}) \quad \bullet \quad P\blacksquare(S_T \geq \text{Goal}) = (1 / M) \sum \blacksquare_{\{S_T \geq \text{Goal}\}}$$

Central Limit Theorem ensures convergence precision narrows to ±1% across M = 10,000 simulated paths.

Ensemble Convergence

- Vectorized simulation of 10,000+ paths in NumPy.
- SLLN: sample mean converges almost surely to true expectation.
- Yields complete empirical probability density function.
- Replaces blind single-point estimates with full distribution.

Downside Tail Risk Metrics

- **Goal Attainment Probability:** exact odds of hitting target.
- **Value at Risk (VaR 95%):** maximum loss at 95% confidence.
- **Conditional VaR (CVaR):** expected loss in worst 5% tail.
- Democratizes institutional hedge-fund risk metrics.

System Architecture & Mathematical Verification

Zero-black-box design, modular computation pipeline, and strict floating-point validation

Three-Tier Architecture

- **Mathematical Engine (models/):** pure functional routines.
- **Verification Suite (tests/):** automated pytest validation.
- **Interactive UI (app.py):** Streamlit parameters & charts.
- **Zero Black Boxes:** every algorithm derived from first principles.

Verification Methodology

- **Excel Parity:** verified against PMT, PV, IRR to $\varepsilon \leq 10^{-7}$.
- **Boundary Tests:** verified at limits $r \to 0$, $t \to 0$, and $m \to \infty$.
- **Domain Guardrails:** strict ValueError on invalid inputs.
- **Analytical Benchmarks:** validated against 24 hand-solved cases.

$| \text{Computer Estimate} - \text{Analytical Truth} | \leq 10^{-7}$ • Automated Pytest Suite Passing

Mathematical reproducibility standard: rigorous verification across all parameter boundaries.

Academic Contributions & Societal Impact

Completed milestones, consumer protection relevance, and open-source availability

Completed Research Milestones

- **12-Volume Reference Monograph (56 Pages):**
- Complete mathematical derivations compiled in theory_and_derivations/.
- Covers discrete accumulation to Itô calculus and Monte Carlo.
- Includes 24 step-by-step hand-calculated case studies.

Consumer Protection Impact

- **Real-World Societal Benefits:**
- Prevents retail citizens from falling into predatory debt traps.
- Eliminates optimistic return illusions via volatility drag modeling.
- Open-source on GitHub under MIT License for public verification.

Thank You! | Open for Questions & Discussion

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